

Evaluating Large Language Models on Aerospace Medicine Principles

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Abstract

Introduction—Large language models (LLMs) hold immense potential to serve as clinical decision-support tools for Earth-independent medical operations. However, the generation of incorrect information may be misleading or even harmful when applied to care in this setting.

Method—To better understand this risk, this work tested two publicly available LLMs, ChatGPT-4 and Google Gemini Advanced (1.0 Ultra), as well as a custom Retrieval-Augmented Generation (RAG) LLM on factual knowledge and clinical reasoning in accordance with published material in aerospace medicine. We also evaluated the consistency of the two public LLMs when answering self-generated board-style questions.

Results—When queried with 857 free-response questions from *Aerospace Medicine Boards Questions and Answers*, ChatGPT-4 had a mean reader score from 4.23 to 5.00 (Likert scale of 1–5) across chapters, whereas Gemini Advanced and the RAG LLM scored 3.30 to 4.91 and 4.69 to 5.00, respectively. When queried with 20 multiple-choice aerospace medicine board questions provided by the American College of Preventive Medicine, ChatGPT-4 and Gemini Advanced responded correctly 70% and 55% of the time, respectively, while the RAG LLM answered 85% correctly. Despite this quantitative measure of high performance, the LLMs tested still exhibited gaps in factual knowledge that potentially could be harmful, a degree of clinical reasoning that may not pass the aerospace medicine board exam, and some inconsistency when answering self-generated questions.

Conclusion—There is considerable promise for LLM use in autonomous medical operations in spaceflight given the anticipated continued rapid pace of development, including advancements in model training, data quality, and fine-tuning methods.

Keywords

aerospace medicine, artificial intelligence, ChatGPT-4, Google Gemini Advanced, Retrieval-Augmented Generation

Introduction

A paradigm of Earth-independent medical operations is fundamental to sustained planetary exploration.¹ Offline-capable multimodal generative artificial intelligence (AI) tools could serve as a highly valuable low-cost and lightweight medical reference tool in supporting Earth-independent medical operations.² However, these AI tools largely rely on large learning models (LLMs), which can generate incorrect information^{3,4} that can be misleading or even harmful when applied to medical care, particularly in remote environments. The objective of this study was to test the ability of 2 popular LLMs, ChatGPT-4 (GPT-4) and Google Gemini Advanced 1.0 Ultra (GA) as well as a custom Retrieval-Augmented Generation (RAG) LLM on factual knowledge and clinical reasoning relative to published material in aerospace medicine.

Medical support for remote launch, landing, and inflight care of astronauts is a highly desirable capability for long-duration space missions.¹ However, medical support from Earth can be limited by communication delays as distance

increases, which can lead to a shift toward store-and-forward telemedicine when immediate communication with flight surgeons is not feasible.⁵ This change in operational paradigm can affect the timely diagnosis and treatment of medical conditions, potentially impacting clinical outcomes and thereby mission success. Additionally, the loss of resupply

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capabilities on deep-space missions threatens the availability of essential medical resources, whereas evacuation to definitive care facilities becomes more complex and time consuming on lunar missions and exceedingly difficult during Mars missions. This in total necessitates a reevaluation of medical support strategies, which include AI tools.

LLMs are advanced deep-learning models designed to understand and generate text in a manner similar to human language.⁶ These models, such as those underlying chat interfaces such as GPT-4 and GA, are trained on vast amounts of text data from sources such as books, articles, and websites. LLMs learn statistical relationships between words, phrases, and sentences during training, enabling them to produce coherent and contextually relevant responses when given prompts or queries.⁷ Recently, LLMs have shown success in medical decision making, particularly when tested on medical board questions.⁸ These models can rapidly develop specialized knowledge across various medical disciplines such as emergency medicine,⁹ orthopedic surgery,¹⁰ and radiology¹¹ by analyzing large amounts of medical data.

Although LLMs exhibit remarkable capabilities, challenges such as overfitting and potential biases in training data exist.¹² LLMs are significantly limited by the quality of the datasets used to train them because they engage in unsupervised training, learning word meanings and relationships based on context.¹³ In practice, strategies such as hyperparameter tuning, regularization, and increasing training data size are employed to mitigate overfitting and underfitting issues in LLM training. Understanding the working principles and limitations of LLMs is therefore crucial for discerning the tasks at which they excel and those where their output might be unreliable. Fine-tuning LLMs for specific tasks is essential but can pose challenges in ensuring accuracy and reliability in content generation.

One method of improving LLM performance on a specific task is the use of RAG. RAG is an advanced framework that integrates traditional LLMs with a retrieval-based component, effectively combining the generation capabilities of language models with knowledge-retrieval abilities akin to search engines. This methodology leverages a dense vector retrieval mechanism that fetches relevant documents or database entries in response to a query. The retrieved information is then fed into the LLM to use as context, which synthesizes and incorporates this external knowledge into the generation of responses or grading decisions. Although RAG is generally thought to improve the reliability of an LLM, understanding the limitations of LLMs, both with and without the use of RAG, is paramount prior to its use during long-duration space missions.

The purpose of this study was to test 2 publicly available LLMs, GPT-4 and GA as well as a custom RAG LLM on factual knowledge and clinical reasoning in accordance with published material in aerospace medicine. We also evaluated the self-consistency of GPT-4 and GA when answering self-generated board-style questions. We hypothesized that LLMs queried with published material in aerospace medicine would demonstrate gaps in factual knowledge, clinical reasoning, and answer consistency, which would improve with the use of RAG. Results from this study will aid in evaluating the limitations of LLMs and the effectiveness of RAG in ameliorating those limitations as well as aid future fine-tuning efforts of LLMs for application on long-duration space missions.

Method

This study tested the factual knowledge, clinical reasoning, and answer consistency of GPT-4, GA, and RAG LLMs. The study methodology is presented in [Table 1](#).

Table 1. Study methodology.

Testing category	Testing conditions	ChatGPT-4	Gemini	RAG LLM
Factual knowledge	Query Evaluation Material	857 free response questions 1–5 grade on Likert scale <i>Aerospace Medicine Board Questions and Answers</i> ¹⁴		
Clinical reasoning	Query Evaluation Material	20 boards-style multiple choice questions (A–D) Graded against provided answer key American College of Preventative Medicine (ACPM) board questions on aerospace medicine		
Answer consistency	Query Evaluation Material	290 self-generated questions (A–D) Graded against provided answer ChatGPT or Gemini Advanced generated questions		RAG LLM could not generate questions for evaluation

RAG, Retrieval-Augmented Generation; LLM, large language model.

Note. This table presents the LLMs and testing conditions to evaluate factual knowledge, clinical reasoning, and answer consistency. The testing conditions are further specified by what type of query was used, how the response was evaluated, and what material was used to generate the query.

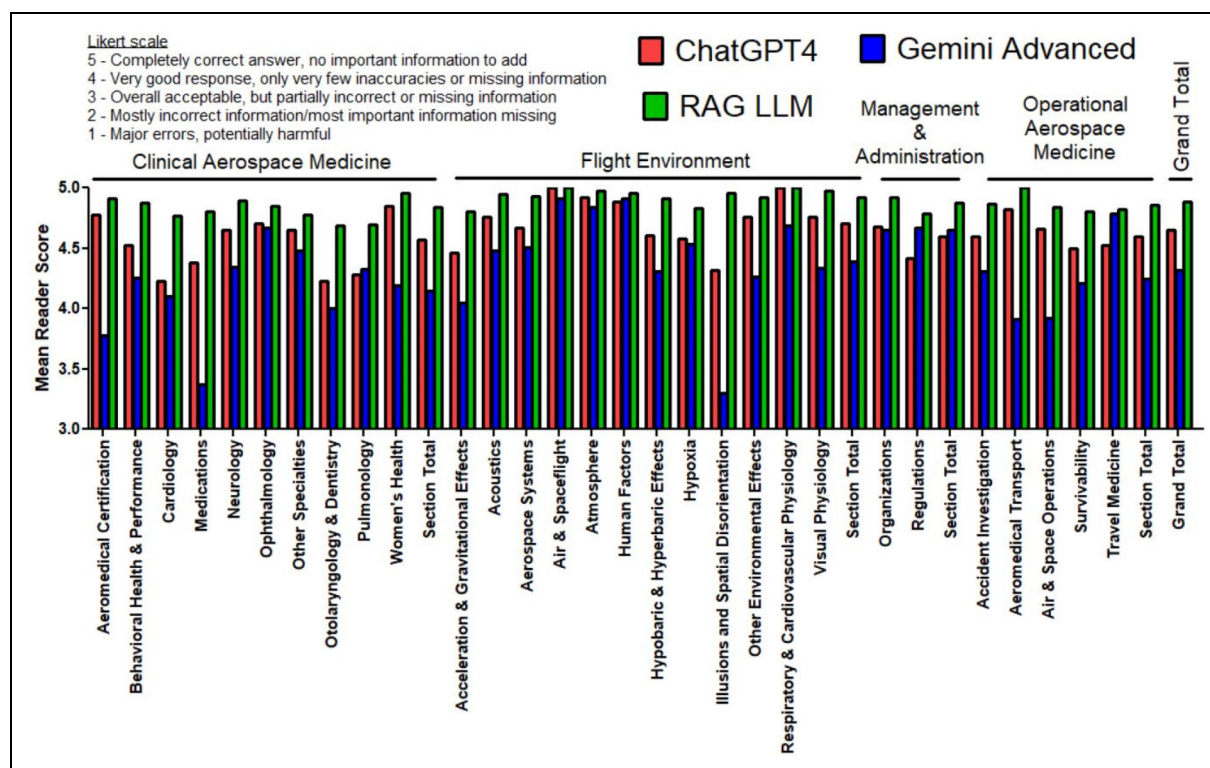


Figure 1. Evaluating clinical reasoning of large language models (LLMs). Performance of ChatGPT-4, Gemini Advanced, and the Retrieval-Augmented Generation (RAG) LLM on aerospace medicine board questions from the American College of Preventive Medicine based on cited resource. This figure quantifies the performance of these LLMs in answering board examination questions from the American College of Preventive Medicine in the field of aerospace medicine. Each response has been evaluated against the provided answer key and scored accordingly to assess the model's effectiveness in clinical reasoning. The bars represent the number of correct responses. The total number of questions is listed below the cited resource.

Testing Factual Knowledge

To test factual knowledge, GPT-4, GA, and the custom RAG LLM were tasked with answering 857 free-response questions from *Aerospace Medicine Boards Questions and Answers*.¹⁴ Prior to the query, GPT-4 was given the following custom instruction: "We are researchers assessing ChatGPT's ability to answer questions about aerospace medicine. Answers should be succinct and no more than a couple sentences. Do not include disclaimers. Assume that your audience is familiar with the material. If an answer is not known, say so." When querying with GA, the prompt began with the following instruction: "We are researchers assessing Gemini's ability to answer questions about aerospace medicine. Answers should be succinct and no more than a couple sentences. Do not include disclaimers. Assume that your audience is familiar with the material. If an answer is not known, say so. Please do not include images and links in responses. Give responses without bullet points or multiple paragraphs." Up to 10 questions were asked within each prompt, and the same chat was used for multiple responses. The RAG LLM was generated first by training on publicly available aerospace medicine

material and then batch queried through a custom application programming interface (API). Responses were then independently graded by 2 readers based on a previously published five-point scale¹⁵ (Figure 1). Mean responses to factual knowledge questions from Reader 1 and Reader 2 were calculated using Microsoft Excel and are presented in Table 2. A 2-way random intraclass correlation coefficient (ICC), calculated using SPSS (SPSS Inc, Chicago, IL), and score distribution are also presented in Table 2. The average scores according to chapter of GPT-4, GA, and RAG LLM are presented in Figure 2. The distributions of answers from GPT-4, GA, and RAG LLM are presented in online Supplemental Tables S1 to S3, respectively.

Testing Clinical Reasoning

To test clinical reasoning, GPT-4, GA, and the RAG LLM were tasked with answering 20 aerospace medicine multiple-choice (choices A–D) practice board questions provided by the American College of Preventive Medicine (ACPM). Prior to the query, GPT-4 and GA were given the following instruction: "Select the single most correct

Table 2. Summary of large language model (LLM) performance on factual knowledge, clinical reasoning, and answer consistency.

Testing category	Measure		ChatGPT-4					Gemini					RAG LLM				
Factual knowledge	Average reader score (Likert scale)		4.65/5 = 93%					4.32/5 = ~86%					4.89/5 = ~98%				
	Score distribution	Grade Count	1 32	2 56	3 73	4 150	5 1403	1 77	2 115	3 166	4 176	5 1180	1 62	2 77	3 120	4 123	5 1332
	ICC (95% CI)		0.750 (0.714–0.781)					0.785 (0.754–0.812)					0.835 (0.812–0.856)				
Clinical reasoning	Graded response		14/20 = 70%					11/20 = 55%					17/20 = 85%				
Answer consistency	Graded response		277/290 = ~96%					286/290 = ~99%					N/A				

RAG, Retrieval-Augmented Generation; LLM, large language model.

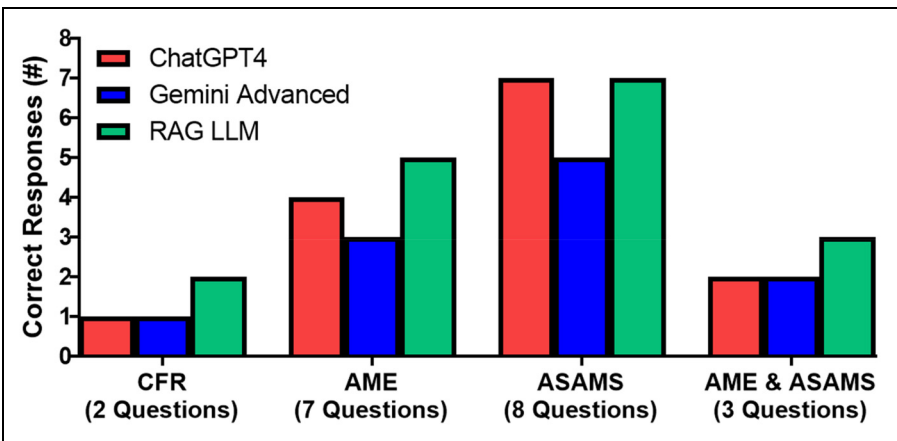


Figure 2. Evaluating factual knowledge of large language models (LLMs). Mean reader scores of ChatGPT-4, Gemini Advanced, and the Retrieval-Augmented Generation (RAG) LLM in answering free-response questions from *Aerospace Medicine Boards Questions and Answers*.¹⁴ Each model's performance is rated based on the accuracy and relevance of its answers. The mean scores are derived from a Likert-scale assessment, providing an empirical measure of each model's capability in generating contextually appropriate and technically accurate responses.

answer to the following question.” Similarly, the RAG LLM was batch queried using a custom application programming interface. Responses from GPT-4, GA, and the RAG LLM were then graded as correct or incorrect, totaled, and presented in Table 2 and presented according to resource cited in Figure 1.

Testing Answer Consistency

GPT-4 and GA were then each tasked with generating 10 multiple-choice questions and answers (A–D) for each chapter title used in *Aerospace Medicine Boards Questions and Answers*.¹⁴ The prompt instruction was: “Generate 10 multiple-choice questions and answers (A–D) for the aerospace medicine board examination on the topic of *insert heading* (eg, hypoxia, human factors, etc). Questions must only have a single correct answer.” The resulting 290 questions generated by GPT-4 were then used as a prompt for GPT-4 to test for answer consistency. Similarly, the resulting

290 questions generated by GA were then used as a prompt for GA. The performance of GPT-4 on GPT-4-generated board-style questions and GA on GA-generated board-style questions are presented in Table 2 and presented according to chapter in Figure 3. The RAG LLM was incapable of question generation and so was unable to be assessed for answer consistency with this method. The 580 multiple-choice questions generated by GPT-4 and GA collectively are presented in online Supplemental Tables S4 through S61.

Results

Performance of LLMs on Aerospace Medicine Principles

Factual Knowledge. The average reader scores of the LLMs on *Aerospace Medicine Boards Questions and Answers*¹⁴ were 4.65, 4.32, and 4.89 out of 5 for GPT-4, GA, and the RAG LLM, respectively. The distribution of responses

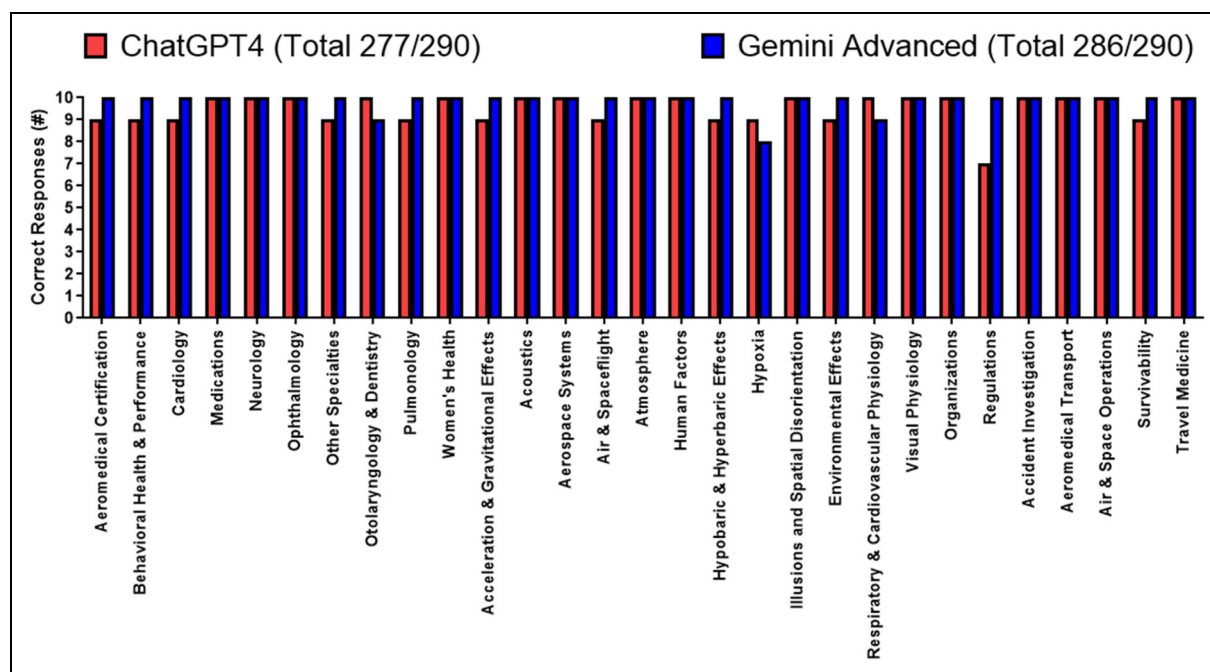


Figure 3. Evaluating answer consistency of large language models (LLMs). Performance of ChatGPT-4 on ChatGPT-4-generated board style questions and Gemini Advanced on advanced generated board style questions. This figure evaluates the performance of ChatGPT-4 and Gemini Advanced on a series of board-style questions that were self-generated by the respective LLM. This self-assessment approach is designed to explore the model's answer consistency. Each response has been evaluated against the answer provided by the LLM and scored accordingly. The bars represent the number of correct responses on 10 questions per listed topic.

graded 1 on the Likert scale (denoting “major errors, potentially harmful”) were 32, 77, and 62 for GPT-4, GA, and the RAG LLM, respectively. The number of responses graded 5 on the Likert scale (denoting “a completely correct answer, no important information to add”) were 1403, 1180, and 1332 for GPT-4, GA, and the RAG LLM, respectively. The ICCs were 0.750 (95% CI, 0.714–0.781) for GPT-4, 0.785 (95% CI, 0.754–0.812) for GA, and 0.835 (95% CI, 0.812–0.856) for the RAG LLM.

Clinical Reasoning. GPT-4 correctly answered 14 of 20 total ACPM aerospace medicine board questions, whereas GA and the RAG LLM answered 11 and 17 of 20, respectively.

Answer Consistency. When queried with GPT-4-generated board-style questions, GPT-4 answered 277 of 290 correctly (Figure 3). When queried with GA-generated board-style questions, GA answered 286 of 290 correctly. A summary of the performance of the LLMs in factual knowledge, clinical reasoning, and answer consistency is presented in Table 1.

Detailed results of the chapter-based performance on factual knowledge, resource-based performance on clinical reasoning, and topic-based performance on answer consistency are presented in Figures 1 through 3, respectively.

Chapter-Based Performance of LLMs on Factual Knowledge

For individual chapters, the mean reader scores for GPT-4 ranged from 4.23 to 5.00 (Likert scale of 1–5; Figure 2). The lowest-scoring chapters were Cardiology, Otolaryngology, and Dentistry, whereas the highest-performing chapters were Air & Spaceflight and Respiratory and Cardiovascular Physiology. GA had a mean reader score range from 3.30 to 4.91 across chapters. The lowest-scoring chapters were Illusions and Spatial Disorientation, while the highest-performing chapter was Human Factors. The RAG LLM had a mean score range of 4.69 to 5.00 across chapters, with Otolaryngology and Dentistry as the lowest scores and Respiratory and Cardiovascular Physiology as the highest scores.

Resource-Based Performance of LLMs on Clinical Reasoning

GPT-4 and GA responded incorrectly to 1 of 2 questions based on the *Code of Federal Regulations* (CFR), whereas the RAG LLM had no incorrect responses. GPT-4 and GA had 3 and 4 incorrect responses, respectively, whereas the RAG LLM had 2 incorrect responses on questions based on the *FAA Guide for Aviation Medical Examiners*. GA had 3 incorrect responses when answering questions base

on the practice guidelines of American Society of Aerospace Medicine Specialists (ASAMS), while GPT-4 and the RAG LLM had only 1 incorrect response. Lastly, when answering questions that cited both the *FAA Guide for Aviation Medical Examiners* and the ASAMS practice guidelines, GPT-4 and GA each had 1 incorrect response, and the RAG LLM had no incorrect responses (Figure 1).

Topic-Based Performance on Answer Consistency

GPT-4 had 1 incorrect response out of 10 total questions in the categories of acceleration and gravitational effects, hypobaric and hyperbaric effects, hypoxia, environmental effects, aeromedical certification, pulmonology, cardiology, behavioral health and performance, other specialties (dermatology, endocrinology, infectious disease, and psychology), and survivability. GPT-4 also had 3 incorrect responses out of 10 in regulations. GA answered 2 incorrectly in hypoxia and only 1 incorrectly in respiratory and cardiovascular physiology and pulmonology, respectively (Figure 3).

Discussion

The purpose of this study was to test the factual knowledge, clinical reasoning, and answer consistency of 2 popular LLMs (GPT-4 and GA) and a custom RAG LLM on material relevant to aerospace medicine. To the best of our knowledge, this study is the first to comprehensively test LLMs on aerospace medicine principles. Regarding factual knowledge, GPT-4 outperformed GA with a mean reader score of 4.65 compared with 4.32 when tested on 857 free-response questions but was surpassed by the custom RAG LLM with a score of 4.89. Interestingly, in terms of the agreement between raters on responses, GA outperformed GPT-4 with ICCs of 0.785 and 0.750 for GA and GPT-4, respectively. However, the RAG LLM again outperformed both GPT-4 and GA with an ICC of 0.835, suggesting strong agreement between the 2 independent readers. In terms of clinical reasoning, GPT-4 again outperformed GA, which correctly answered 14 compared with 11 out of the 20 ACPM aerospace medicine board practice questions. This was once more surpassed by the custom RAG LLM with a score of 17 of 20. However, GPT-4 demonstrated poorer answer consistency (~95%) when responding to its own board-style questions, while GA had a greater answer consistency of (~99%) when responding to questions generated by GA. The hypothesis of this study was supported because the general LLMs still demonstrated remaining gaps in factual knowledge, clinical reasoning, and answer consistency that were improved with the use of RAG.

*Aerospace Medicine Boards Questions and Answers*¹⁴ is a publicly available resource with the primary use of assisting with preparation for the aerospace medicine board examination. The book covers essential topics from a wide

variety of disciplines, such as clinical aerospace medicine, the flight environment, management and administration, and operational aerospace medicine. Individuals studying board-style questions engage in critical thinking, analysis, and application of knowledge to enhance their understanding of the subject matter. Much like human learners, LLMs benefit from systematic acquisition of knowledge through engagement with questions and responses.^{16,17} Both processes emphasize the importance of active engagement with the material to foster a deeper comprehension of the subject matter and can identify potential gaps in knowledge.

When GPT-4 and GA were presented with questions from *Aerospace Medicine Boards Questions and Answers*,¹⁴ both LLMs had an average response between 4 and 5 on the Likert scale (4.65 and 4.32, respectively). Scoring between 4 and 5 on a Likert scale indicates a positive response or agreement with the statements presented (Figure 2). In the context of the *Aerospace Medicine Boards Questions and Answers*, this scoring range would suggest a strong level of agreement or satisfaction with the content of the LLM answers by the human rater, potentially reflecting a high level of LLM understanding and confidence in the material covered in the questions and answers provided. However, there were 32 responses by GPT-4 and 77 responses by GA that were graded as 1 on the Likert scale (Table 2). These denote major errors in understanding that may be potentially harmful in clinical use. Although a total of 110 instances across 1714 graded responses equates to only ~6%, the severity of error paired with no lack of confidence as one might sometimes expect from a human response in the setting of a knowledge gap may have serious implications to the patient, spacecraft, and mission in an operational setting. Similarly, the custom RAG LLM also had an average response between 4 and 5 on the Likert scale, but interestingly, it had a higher number of responses graded as 1 (Table 2) than GPT-4. This suggests that the mean accuracy of the responses was superior but that potentially dangerous responses were not reduced.

The performances of the LLMs on 20 aerospace medicine practice board questions from the ACPM were 70%, 55%, and 85% for GPT-4, GA, and the RAG LLM, respectively. Although the 20 questions provided by the ACPM were not intended to predict exam performance, the passing score on the aerospace medicine board examination is 60%.¹⁸ Since GPT-4 launched in early 2023, it has been tested on multiple medical board exams. GPT-4 performed well on US Medical Licensing Exam Step 1, Step 2 CK, and Step 3 practice questions, answering a high proportion of questions correctly (88%, 86%, and 90%, respectively).¹⁹ GA is powered by a different underlying LLM, which became publicly available in early 2024 and has not been tested as rigorously on medical licensing exam material. The aerospace medicine practice board questions from the

ACPM used 3 resources: the *FAA Guide for Aviation Medical Examiners*,²⁰ the ASAMS practice guidelines,²¹ and the CFR.²² The *FAA Guide for Aviation Medical Examiners* and ASAMS practice guidelines serve as important references, best practices, and guidelines for medical professionals to maintain optimal health, safety, and performance standards for individuals working in the aerospace and aviation fields. Additionally, the CFR acts as a comprehensive regulatory framework, guiding aviation professionals in adhering to legal standards and ensuring the overall safety and efficiency of aviation operations within the United States.

LLMs have shown the capability to generate board-style questions, but their consistency in answering these questions can vary.²³ When GPT-4 and GA were tasked with generating aerospace medicine-specific board questions, both LLMs exhibited answer consistency >95%. Perhaps unsurprisingly, the chapters where GPT-4 demonstrated some of its greatest limitations in factual knowledge (Cardiology, Pulmonology, and Regulations, the 29th, 27th, and 24th ranked sections out of 29) also were categories in which GPT-4 had answer inconsistency (9 of 10, 9 of 10, and 7 of 10, respectively). Interestingly, GPT-4 still provided an incorrect response in Aeromedical Certification, one of its better-performing chapters in the domain of factual knowledge. GA had only 1 incorrect responses in Respiratory and Cardiovascular Physiology, Hypoxia, and Pulmonology, its fifth, ninth, and 15th highest-performing categories in factual knowledge, respectively. Taken together, the tested LLMs demonstrated a lack of question and answer fidelity that would limit trustworthiness in providing accurate clinical decision-making support.

In this study, we demonstrate that LLMs exhibit limitations in factual knowledge, clinical reasoning, and answer consistency that impact their reliability in applications in aerospace medicine. The LLMs we tested exhibited gaps in factual knowledge that potentially could be harmful, clinical reasoning that may not pass the aerospace medicine board exam, and inconsistency when answering their own questions. Nevertheless, these LLMs demonstrated impressive performance on free-response questions and tremendous promise given their rapid pace of development, including advancements in model training, data quality, and fine-tuning methods. These techniques are important to enhance their performance and reliability in generating high-quality responses that could be suitable for aerospace medicine applications.

Limitations

Limitations of this study include but are not restricted to limited publicly available testing materials applicable to aerospace medicine, testing material available only in a traditional examination format, and lack of assessment of LLM-generated questions for quality. The evaluation of

LLM performance in clinical scenarios benchmarked against accepted aerospace medicine operational guidelines and workflows would provide even greater insight to their accuracy in this domain. Written examinations are also often an unrepresentative measure of clinical capability due to their limited ability to assess practical skills and real-world decision making. Lastly, further assessment of the quality of LLM-generated board-style questions may enable the rapid and low-cost generation of high-quality educational and assessment material for the field.

Limitations of LLM application to aerospace medicine more broadly are also important to mention. LLMs can suffer from hallucinations, generating false or logically incorrect text that appears plausible.²⁴ To address this, methods such as prompt chaining and fact checking are being explored to improve the accuracy of information provided by LLMs. Although LLMs show potential in medical education by assisting in clinical reasoning and knowledge acquisition, hurdles remain. One concern is that LLMs may reflect outdated scientific consensus instead of current knowledge, highlighting the need for continual learning and retrieval from evolving sources. Inconsistency is another challenge with LLMs, where small changes in prompts could lead to divergent responses, potentially affecting the model's response to free-response and multiple-choice questions. LLMs lack the ability to discern true information from false within their datasets, leading to instances where they perpetuate incorrect information until corrected by humans. These models store facts implicitly in their parameters, which can cause inaccuracies or deviations from the truth in their generated content. Despite challenges such as overfitting, high fine-tuning overhead, and high inference latency, LLMs have shown promise with commendable accuracy levels when properly tuned and refined.

Furthermore, operational limitations also exist, such as the computer power required to locally host an LLM on a craft during a long-duration mission. First, the lack of radiation-hardened hardware suitable for space missions can hinder the deployment of LLMs in such environments, where exposure to cosmic radiation can affect sensitive electronic components. Second, the rapid pace of development in LLM technology may lead to compatibility issues with existing space systems and infrastructure, making it challenging to integrate these advanced models seamlessly into the existing architecture. Additionally, the vast majority of scientific literature available for LLM training is based on terrestrial physiology rather than physiology in the reduced-gravity environment of spaceflight, which could limit their applicability in addressing specific medical emergencies that astronauts may encounter in space. Although the RAG LLM used in this study was the first LLM, to our knowledge, to be specifically trained on aerospace medicine materials, additional normative data to the spaceflight environment likely would result in even greater improvement of LLM performance on aerospace medicine principles.

In terms of application to aerospace medicine, it is the opinion of these authors that LLMs should not be intended to supplant the crew medical officer but rather serve as a clinical decision-support tool once sufficiently developed and rigorously tested for this use. LLMs no doubt will play a crucial role in assisting terrestrial medical professionals by providing valuable insights and information to aid in decision-making processes. The use of LLMs as clinical decision-support tools offers benefits such as expanding differential diagnoses, retrieving clinical guidelines, and delineating treatment strategies. However, medical operations teams no doubt will face challenges implementing these tools into existing workflows, particularly in the near term, when using LLMs that lack multimodal capabilities in settings that require visual input and significant manual skill in exam and procedural performance.

Despite these challenges, future directions for investigations in this area are numerous. For example, multimodal LLMs and other AI tools may be able generate just-in-time training and procedural guidance in a variety of formats, including audiovisual, given recent developments in text to video, such as Sora (OpenAI, San Francisco, CA). Additionally, with the rise of custom GPTs, tailored versions of these broadly available LLMs may be trained for the specific population and pathology expected in spaceflight, particularly on the text documents containing the specific recommendations on which medical advice should be provided. Overall, although LLMs offer significant potential in aerospace medicine, they are for now best used as tools to complement human expertise rather than replace the critical clinical reasoning abilities that are essential for making complex medical decisions, particularly those influenced by individual patient needs and nuanced clinical contexts. The results of this study demonstrate that gaps remain in the factual knowledge, clinical reasoning, and answer consistency of these 2 LLMs despite capabilities that are already advanced. The development of future task-specific or fine-tuned LLMs for aerospace medicine, like the RAG LLM used in this study, could further mitigate these knowledge deficiencies and allow for smaller models with resource utilization levels appropriate for a long-duration space mission.

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Author Contribution(s)

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Cole A. Davis: Methodology; Writing – original draft; Writing – review & editing.

Shawn M. Pickett: Formal analysis; Investigation; Methodology; Writing – original draft; Writing – review & editing.

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Supplemental Material

Supplementary material associated with this article can be found in the online version at <https://doi.org/10.1177/10806032251330628>.

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